

COMP66060 Workshop

Research Writing

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Course overview – where are we?

Date	Time	Topic	Lecturer	Location
16 June 2026	09:00-12:00	Project Planning	Ainur Begalinova	Kilburn_TH 1.1
	13:00-16:00	Literature Review	Davide Bertozzi	Kilburn_TH 1.1
17 June 2026	09:00-12:00	Ethics in CS	Ainur Begalinova	Schuster_Rutherford TH
	13:00-16:00	Research Conduct and Evaluation	Davide Bertozzi	Schuster_Rutherford TH
18 June 2026	09:00-12:00	Research Writing	Chenghua Lin	Schuster_Rutherford TH
	13:00-15:00	Responsible Research and Innovation	Prof Mateja Jamnik (University of Cambridge)	Schuster_Rutherford TH

- Today we will cover research writing

Engineers and Good Writing



Vannevar Bush (1890-1974) served as director of the Office of Scientific Research and Development throughout World War II. The nation's top designer of computers in the 1930's.

- Writing is essential to the success of an electrical engineer, Bush wrote in 1922, in his first textbook, "The Principles of Electrical Engineering" (co-authored with a colleague, William H. Timbie). The success of any engineer's plan, Bush insisted, often depends on good writing:
 - *"Once the plan has been decided upon, he must convince his superiors that the plan should be carried out. This convincing requires that the engineer write brief clear English, which adequately and concisely conveys the meaning in a convincing way. Good proposals have been turned down because the engineers who drafted them could not present them in convincing form."*

IEEE Spectrum Mar 2022
 "The Essential Writings of Vannevar Bush"
 Columbia University Press (2022) 3

Motivation – Why is writing important?

- Research writing is critical as it allows you to clearly express your ideas, work and conclusions

Clarity is King

- If the markers cannot understand what you have done - then they cannot give a good mark

Outline

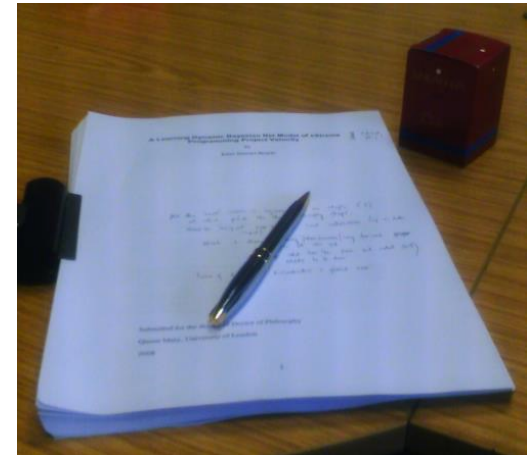
- Useful tips on technical writing
 - Some thoughts on what is expected in creating high quality written work
 - Written English often has no one “right” answer
- Dissertation
 - Structure and sections
 - Referencing
 - Figures and equations



<https://essaytools.com/dissertation-writing-tips/>

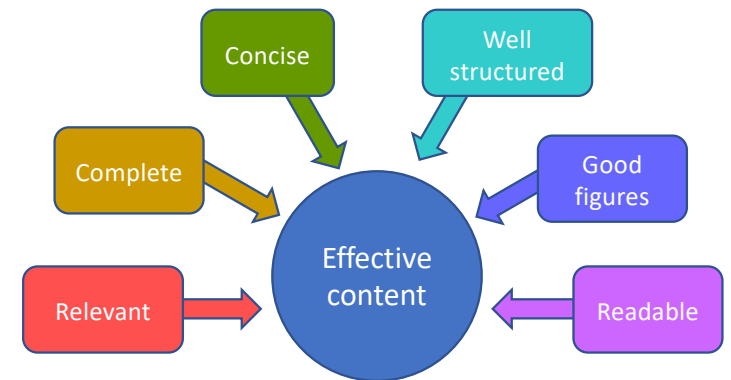
What makes good Technical writing?

- Good technical writing is...
 - Efficient communication of ideas
 - Clear and simple as possible
 - Precise
 - Interesting and engaging
- Good technical writing **is not**...
 - Ambiguous
 - Flowery
 - Filled with unnecessary jargon
 - Boring
- Luckily this is also the easiest kind of writing



Technical Writing: Main points

- 1) Be clear about the aim of your writing
- 2) Be clear about who are the target readers
- 3) Use readable English
- 4) Use appropriate structure
- 5) Use appropriate formatting (REQUIRED)
- 6) Get feedback on your writing



- We now consider these points in more detail

1) What is the aim of your writing?

- In writing your dissertation a major aim is to get a good mark – so look at the marking scheme
- The marking scheme should also help focus you on more specific aims of the report, e.g. to provide an overview of the literature for a general audience
- Describe your research methods, results and conclusions



fa16850733 freeart.com ©

2) Who is it for?

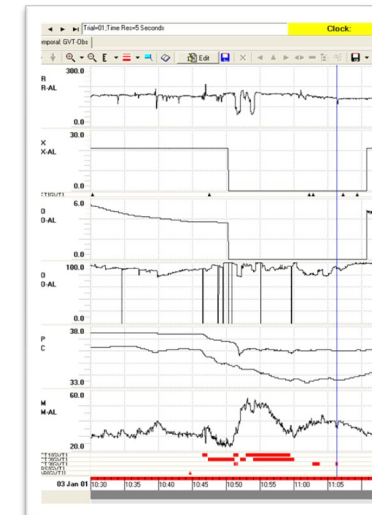
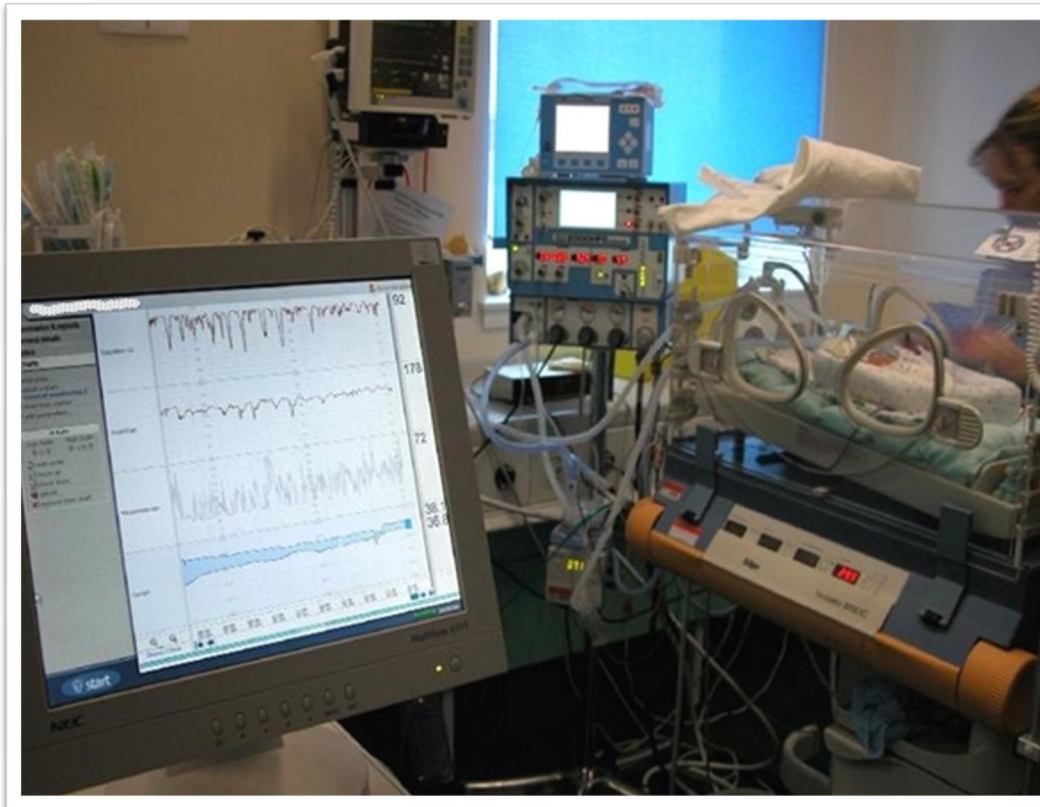


- Your dissertation will be marked by the supervisor and a 2nd marker, but they are not the only target audience
- The report should be accessible to a broader audience, usually your peers
- Assume the reader knows the general field (e.g. CS) but does not know the specifics of your project
- Imagine writing for you before you started work on your project

Example: BabyTalk

- Summarise clinical data about premature babies in neonatal ICU
- Input: sensor data; records of actions and observations by medical staff
- Output: multi-paragraph texts, summarised data for different audiences
- A research project of the University of Aberdeen

Babytalk: Neonatal ICU



BT-Clinicians texts (extract)

- Short summary supporting real-time decision making by clinicians:

By 11:00 the baby had been hand-bagged a number of times causing 2 successive **bradycardias**. She was successfully re-intubated after 2 attempts. The baby was sucked out twice. At 11:02 FIO2 was raised to 79%.

BT-Family text (extract)

- Page-long text for parents:

Yesterday, John was on a ventilator. The mode of ventilation is Bilevel Positive Airway Pressure (BiPAP) Ventilation. This machine helps to provide the support that enables him to breathe more comfortably. Since last week, his inspired Oxygen (FiO₂) was lowered from 56 % to 21 % (which is the same as normal air). This is a positive development for your child.

During the day, Nurse Johnson looked after your baby. Nurse Stevens cared for your baby during the night.

3) Use readable English

- Aim for shorter sentences: avoid too many commas and conjunctions (and, but, yet)
- Use lists and bullet points where appropriate
- Do not use overcomplicated words
- Avoid adding words which add nothing to the meaning
- Minimise style and grammar errors

Use short sentences

- A sentence should contain one unit of information, that is:

one thought or idea

- If you can split a sentence in two without losing the meaning, then do it.
- “Our model is a modification of Switching Linear Dynamical Systems and we show that it performs better at generation than other models.”

Keep it short and sweet to put forth intention without adding confusion	• The topic sentence should act as middle ground • Don't: encompass the info of your whole paragraph • Use topic sentence to set forth action, but don't make the reader hunt down meaning
---	---

<https://www.wikihow.com>

Use lists and bullet points

- Most readers get confused when faced by a long list embedded within prose.
- Algorithms, Recipes, Directions and other Instructions are better written as *lists* or included as *tables*.
- If you notice that you are effectively writing a list, then use a numbered list or bullet points.

However, the parameter estimation procedure also has some disadvantages such as:

- It is more complex to implement and understand.
- The characteristics of the system extracted during the step response system identification procedure give insight about the system.
- Sometimes a PRBS experiment may need to be rerun. This may happen in the case that the value in the PRBS signal is constantly the same, which although is highly unlikely is possible.

Use readable words/expressions

- Avoid unnecessarily complicated words and phrases:

In order to **fully utilise** ion-irradiation to **attune** the magnetic properties of FeRh it is **absolutely essential** to have a detailed **comprehension** of the mechanisms involved.

- Avoid stock phrases:

“There is a reasonable expectation that...”

- Avoid inappropriate use of jargon:

Define acronyms the first time it is used and use a glossary with the definition.



Avoid unnecessary words

- Example 1:
 - “After specification we are in a position to begin a detailed design”
 - “After specification we can begin detailed design”
- Example 2:
 - “This is done by means of inserting an artificial fault”
 - “This is done by inserting an artificial fault”

Edit Ruthlessly

Somebody ~~has~~ said that words are ~~a lot~~ like inflated money - the more ~~of them that~~ you use, the less each one ~~of them~~ is worth. ~~Right on.~~ Go through your entire letter ~~just~~ as many times as it takes. ~~Search out and~~ annihilate all unnecessary words, ~~and~~ sentences—even ~~entire~~ paragraphs.

Malcolm Forbes
("How to write a business letter")

Active and passive styles

- Active: “Joe tested the software”
- Passive: “The software was tested by Joe”

It is the subject or data that is important - not the person

- In scientific writing the passive style is generally considered more formal
- However, the passive style writing can lead to longer and clumsier sentences
- Use the active style only when really needed

Personal style

- An overly personal style is generally to be avoided in scientific writing
- However, some personal phrases can result in a more readable passage and avoids contorted sentence construction! – Keep it simple
- A dissertation is your work not “our” work, so could use:
 - “*I show...*” rather than “*we show..*” to describe what is shown.
 - But much better “***The data show...***”
- But “We see that by replacing...” is OK as the “we” is you and the readers.
- Better still “Replacing xxx leads to...”



Dissertation (4th Sept.)

1. Dissertation

- Approx. 8,000 words
- Submit as a **single PDF file**
- Must follow formatting guidelines and include all required pages

Submission Instructions

- Assessment rubric:  [MSc_Report_and_Video_Rubric.pdf](#) 
- Ensure your submission complies with the **University Presentation of Theses Policy**

Dissertation (4th Sept.): Structure and sections

- Abstract
- Contents
- Numbered Sections
 - 1. Introduction
 - Background (brief literature review)
 - 2. Methodology - Research (or experimental) methods
 - 3. Results and discussion
 - 4. Evaluation and Reflection
 - 5. Conclusion & future work
- List of References



**Remember rules
on plagiarism and
contract cheating!!**

Abstract

- Should summarise the dissertation – it is not an introduction – think “executive summary”
- Should summarise the results and conclusions of the work
- It should be self-contained with no references
- Should be informative (contain results) not just descriptive (say what the report is about)



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Abstract

This thesis presents the development and evaluation of advanced generative models, specifically Variational Autoencoders and Diffusion Models, for the purpose of face image generation. A significant component of the work also involved the construction of an iOS application, serving as a user-friendly interface that allows users to interact with the generative models. The image generation process was improved by incorporating GFPGAN, an image upscaler, to enhance the realism of the produced images. Additionally, latent space arithmetic was applied to condition the Variational Autoencoder models.

The project not only met its objectives by generating highly realistic face images and integrating the models into a functional iOS app but also highlighted the ethical considerations associated with the use of generative AI for face image generation. These include concerns over bias propagation, privacy issues, and the potential for misuse, emphasizing the critical need for an unbiased and representative dataset.

Abstract

Dissertation

Introduction

- Introduction and overview of the dissertation
 - What is it all about?
 - Why does it matter?
 - What is the research area?
- Must include aims and objectives
 - Is there a research question?
- Can include a description of the structure of the dissertation
 - e.g. chapter 2 covers theory...
 - e.g. chapter 3 methods
- Section introductions: if a section has sub-sections then the text before 1st sub-section should be an introduction of that section.



Introduction: Background

- The introduction must include a literature review
 - What is known about the field?
 - Need to set the work in the context of what others have done
 - Not extensive: just enough to understand the problem and justify the approach



Methodology

- Research (or experimental) methods section should be used to describe the tools and techniques employed or developed

How was the project work done?

- In hardware projects this would include instrumentation and experimental methods
- In software projects it could include choice of programming language and software package and methods adopted



Research Methodology

Methodology

- Research (c) be used to employed c

How

- In hardware instrument

- In software projects it could include choice of programming language and software package and methods adopted

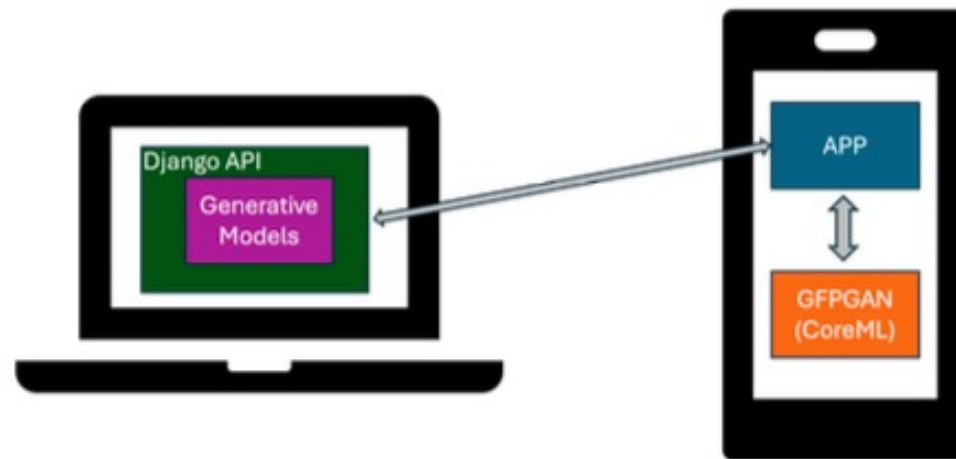


Figure 4.4: The complete architecture



Research Methodology

Results and discussion

- What results were obtained
 - What advances have been made?
 - Discussion of data e.g. test results
 - What information and findings can be deduced from the data
 - Detailed discussion of validity of conclusions
 - What testing and evaluation was done?
 - **Why should we believe the results? Reliability, limitations and uncertainty in the methods/outcomes.**



Conclusion

- More important than you think
 - What have you learnt? (this could be positive or negative)
- Use it to...
 - Give your overall conclusions
 - Reflect on your progress on each objective
 - Demonstrate what you learned
 - Discuss future avenues of investigation

§
KEEP
CALM
AND
FINISH YOUR
DISSERTATION

Reference styles



- Two main styles:
 - Listed numerically in the order of appearance in the manuscript. Cited either like this [1] or like this¹. E.g “Hartley showed that...[1]” or “Hartley¹ showed that...”
 - Listed alphabetically and cited by (name, date) or name (date). E.g. “Hartley (1957) showed that...” or “It was shown that...(Hartley, 1957)”.
- Word and latex will automatically create reference lists in either order – for latex use bibtex and use natbib for (name, date) style


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Computer Science > Computation and Language

[Submitted on 12 Jun 2017 (v1), last revised 2 Aug 2023 (this version, v7)]

Attention Is All You Need

[Ashish Vaswani](#), [Noam Shazeer](#), [Niki Parmar](#), [Jakob Uszkoreit](#), [Llion Jones](#), [Aidan N. Gomez](#), [Lukasz Kaiser](#), [Illia Polosukhin](#)

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model achieves 40.6 BLEU, improving over the existing best results, including ensembles by over 2 BLEU. On eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks like language modeling, automatic machine translation, machine reading comprehension, and named entity recognition.

Comments: 15 pages, 5 figures
 Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)
 Cite as: [arXiv:1706.03762](#) [cs.CL]
 (or [arXiv:1706.03762v7](#) [cs.CL] for this version)
<https://doi.org/10.48550/arXiv.1706.03762>

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[\[v3\]](#) Tue, 20 Jun 2017 05:20:02 UTC (1,125 KB)
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BibTeX formatted citation

```
@misc{vaswani2023attentionneed,
  title={Attention Is All You Need},
  author={Ashish Vaswani and Noam Shazeer and Niki Parmar and Jakob Uszkoreit and Llion Jones and Aidan N. Gomez and Lukasz Kaiser and Illia Polosukhin},
  year={2023},
  eprint={1706.03762},
  archivePrefix={arXiv},
  primaryClass={cs.CL},
  url={https://arxiv.org/abs/1706.03762},
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```

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Dissertations - Correct Formatting

- It is a **requirement** that dissertations be correctly presented

<http://documents.manchester.ac.uk/display.aspx?DocID=2863>

- Section numbers
 - Sufficient to provide a coherent structure
- Figures and Tables
 - DO NOT forget about the captions
- Maths
- References

CONTENTS

1. Introduction
2. General information
3. Formatting
4. Required Pages
5. Sample Title Page
6. Dissertation Submission Checklist

<http://documents.manchester.ac.uk/display.aspx?DocID=2863>

Section numbering

- Sections at the same level should be similar in length
 - Introduction and Conclusion may be somewhat shorter
- Use sub-sections (1.2) and sub-sub-sections (1.2.1) but avoid sub-sub-sub-sections (1.2.1.2).
- Use your word processor to remember the section numbers when referencing (use `\label{key}` and `\ref{key}` in latex)

List of Figures	3
List of Tables	7
Abstract	8
Declaration	9
Intellectual Property Statement	10
Acknowledgements	11
1. Introduction	12
1.1 Aim and Objectives	12
1.2 Project Context	13
1.3 Deliverables	13
1.4 Dissertation structure	14
2. Background	15
2.1 Control Theory	15
2.1.1 Transfer function	16
2.1.2 Continuous time and discrete time	18
2.1.3 s and z domains	18
2.1.4 Deviation variables	19
2.2 PID control	20
2.2.1 PID Tuning techniques	22
2.3 Hardware components and software tools	25
2.3.1 Raspberry Pi	25
2.3.2 Python programming language	26
2.3.3 Input and output resolution	27

Figures and Tables

- A picture paints a 1000 words!
- Good figures are a key part of your written work:
 - Always include an informative caption.
 - Always integrate the figure in the text.
- Number by section in a dissertation, e.g. Figure 5.2 is 2nd figure in Section 5.
- Use your word processor to remember Figure numbers (use `\label{key}` and `\ref{key}` in latex).



Mona Lisa - Wikipedia

Figure example

- The figure and caption are self-contained
- It is possible to gain some understanding what the figure is trying to show without reference to the text.

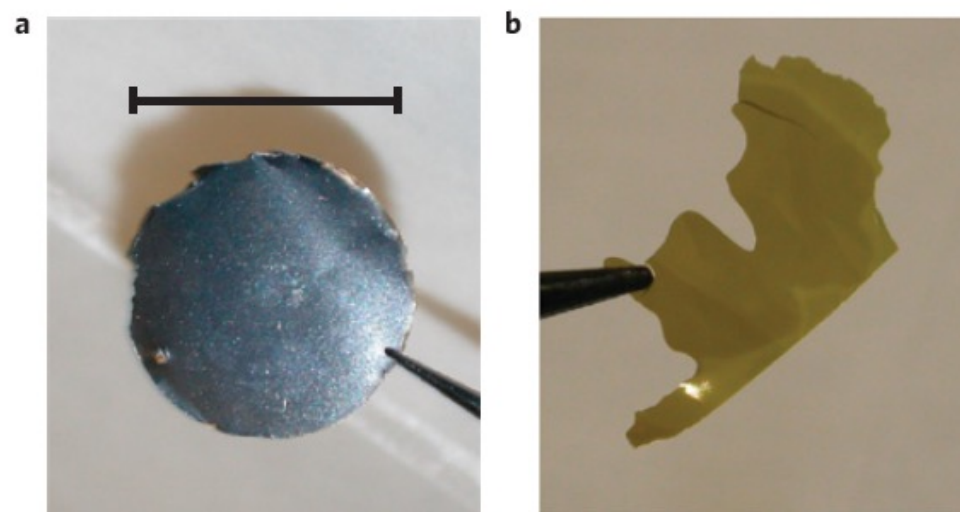
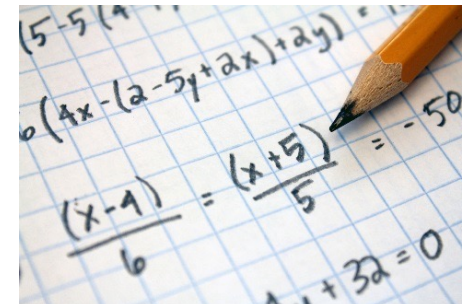


Figure 1 | Experimental samples. a,b, Optical images of starting (**a**) and fully fluorinated (**b**) graphene laminates. The scale bar is 2 cm. The laminates consist of 10–50 nm graphene crystallites, predominantly mono- and bilayers¹⁵, aligned parallel to each other (as seen in a scanning electron microscope), rotationally disordered and, consequently, electronically decoupled²².

Nair et al. Nature Physics PUBLISHED ONLINE: 10 JANUARY 2012 | DOI: 10.1038/NPHYS2183

Maths



<https://www.japantimes.co.jp/>

- Use different fonts for maths, either *italic* or **bold**. This is important within text, e.g. “where α refers to amplitude”.
- Anything longer than $x + y = z$ should be given its own line.
 - **Number every equation**
- Punctuate maths just like English. Write it in the way you read it.
- Use proper maths objects. If you have a lot of maths then consider using latex

References



<http://guides.library.uwa.edu.au/referencinguwa>

- Styles vary but references must contain:
 - Journal articles: Authors, Title, Journal, Volume number, Page range, Date.
 - Books: Authors, Title, Edition, Publisher, Date.
 - Conference articles: Authors, Title and either Conference Title, (Place, Date) or Volume Title, Editors (Publisher, Date).
- Easiest to look at examples in journals, theses and dissertations to see what is acceptable.

Dissertation templates - Latex

- The computer science dissertation template is available at:

<https://studentnet.cs.manchester.ac.uk/resources/latex/MUThesis/>

- Colleagues in EEE have an overleaf template which you are free to use but MUST be customized for computer science:

<https://www.overleaf.com/latex/templates/msc-dissertation-template-the-university-of-manchester-eee/wxtvdqdhnpwp>

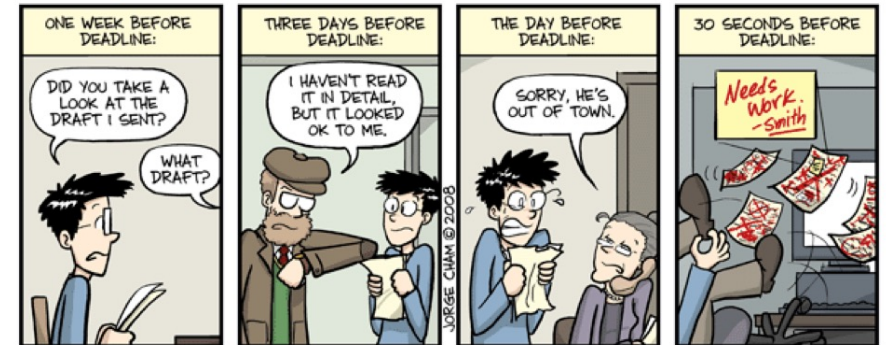
- Responsibility for ensuring compliance with the University of Manchester Presentation of Theses Policy remains with the candidate

Getting feedback

- Early on, get feedback on the overall structure of your report/dissertation
- Get feedback on individual sections initially. Leave enough time for this
- **Important:** Only get feedback on text that follows the rules here and has been spell checked. Otherwise your supervisor is just an expensive spell checker.
- Consider peer-reviewing among yourselves.

Piled Higher and Deeper by Jorge Cham

www.phdcomics.com



title: "Needs work" - originally published 4/9/2008

WWW.PHDCOMICS.COM

and Finally...

- Questions?

Exercise

Write down the aim and objectives of your project

- **Aim:** A broad statement of the overall intended outcome of the project. One sentence only.
- **Objectives:** Specific, measurable steps that will help achieve the aim. Typically 3–5 items.

Example

- **Aim:**

- To develop and evaluate a machine learning model for accurately predicting customer churn in a subscription-based service.

- **Objectives:**

- To conduct a literature review on existing churn prediction models and key influencing factors.
- To preprocess and analyze a real-world customer dataset to identify relevant features.
- To design and implement multiple machine learning models (e.g., logistic regression, random forest, XGBoost) for churn prediction.
- To evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- To interpret results and recommend customer retention strategies.